

KOOPMAN INVARIANCE OF GAUSSIAN RKHSs FOR LINEAR STOCHASTIC DIFFERENTIAL EQUATIONS: A NECESSARY AND SUFFICIENT CONDITION

ABSTRACT. Philipp, Schaller, Worthmann, Peitz and Nüske showed that the reproducing kernel Hilbert space H_C of the Gaussian kernel $\exp(-(x-y)^\top C^{-2}(x-y))$ is invariant under the Koopman semigroup of the linear stochastic differential equation $dX_t = AX_t dt + B dW_t$, with A Hurwitz and (A, B) controllable, whenever $\frac{1}{2}(AC^2 + C^2A^\top) \leq BB^\top$. They asked whether this condition is also necessary. We show that it is not, and we identify the exact condition: H_C is invariant under every Koopman operator K^t , $t \geq 0$, if and only if $\frac{1}{2}(AC^2 + C^2A^\top) \leq 2BB^\top$. For a single time t we prove that K^t maps H_C isomorphically onto the Gaussian RKHS with matrix $(e^{-At}(C^2 + 4\Sigma(t))e^{-A^\top t})^{1/2}$, where $\Sigma(t)$ is the conditional covariance of X_t given X_0 . Consequently $K^t H_C \subset H_C$ holds if and only if $e^{At}C^2e^{A^\top t} \leq C^2 + 4\Sigma(t)$, and then $\|K^t\| = e^{-t \operatorname{Tr} A/2}$. For Hurwitz A , invariance for all $t \geq 0$ holds if and only if the ellipsoid of $C^2 + 4\Sigma$, where Σ is the stationary covariance, is forward invariant under $\dot{x} = Ax$. The proofs rest on a Fourier description of Gaussian RKHSs, in which K^t becomes a weighted composition operator.

1. INTRODUCTION

The Koopman operator was introduced in [7] for deterministic systems. For a stochastic dynamical system $(X_t)_{t \geq 0}$ it acts on observables f by

$$(K^t f)(x) = \mathbb{E}[f(X_t) \mid X_0 = x].$$

It is linear even when the dynamics are not, and it can be approximated from data, for instance by extended dynamic mode decomposition (EDMD) [17] and its kernel-based variants [5, 18]; see [9] for an overview. As pointed out in [12], the derivation of error bounds for kernel-based approximations of Koopman operators [3, 6, 8, 11] benefits greatly from the reproducing kernel Hilbert space (RKHS) of the kernel being invariant under the Koopman operator. This raises the question of which kernels and which systems are compatible in this sense.

Philipp, Schaller, Worthmann, Peitz and Nüske [12] studied this question for the linear stochastic differential equation

$$(1.1) \quad dX_t = AX_t dt + B dW_t,$$

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where $A \in \mathbb{R}^{d \times d}$, $B \in \mathbb{R}^{d \times m}$ and W is an m -dimensional Brownian motion, together with the Gaussian kernel

$$k^C(x, y) = \exp(-(x - y)^\top C^{-2}(x - y)) = \exp(-|C^{-1}(x - y)|^2), \quad x, y \in \mathbb{R}^d,$$

with $C \in \mathbb{R}^{d \times d}$ symmetric positive definite. Let H_C be the (real) RKHS of k^C on $\mathbb{R}^d$, with norm $\|\cdot\|_C$, and let

$$(1.2) \quad \Sigma(t) := \int_0^t e^{As} B B^\top e^{A^\top s} ds \quad (t \geq 0)$$

be the covariance of X_t given X_0 . Under the standing assumptions of [12], namely $m \leq d$, (A, B) controllable and A Hurwitz, they proved [12, Theorem 2.1] that the Lyapunov-type inequality

$$(1.3) \quad \frac{1}{2}(AC^2 + C^2A^\top) \leq BB^\top$$

(in the Loewner order; this is [12, (2.3)]) implies $K^t H_C \subset H_C$ for all $t \geq 0$, with $\|K^t f\|_C \leq e^{-t \operatorname{Tr} A/2} \|f\|_C$. Whether (1.3) is also necessary was left open [12, Remark 2.2(c) and Remark 4.2].

Main result. The answer is negative. The following theorem characterizes invariance completely, both at a fixed time and for the whole semigroup. It needs none of the standing assumptions of [12].

Theorem 1.1. *Let $A \in \mathbb{R}^{d \times d}$ and $B \in \mathbb{R}^{d \times m}$ be arbitrary, and let $C \in \mathbb{R}^{d \times d}$ be symmetric positive definite.*

(a) *For every $t \geq 0$,*

$$(1.4) \quad K^t H_C \subset H_C \iff e^{At} C^2 e^{A^\top t} \leq C^2 + 4\Sigma(t),$$

and in this case $\|K^t\|_{H_C \rightarrow H_C} = e^{-t \operatorname{Tr} A/2}$.

(b) *$K^t H_C \subset H_C$ holds for every $t \geq 0$ if and only if*

$$(1.5) \quad \frac{1}{2}(AC^2 + C^2A^\top) \leq 2BB^\top.$$

If (1.5) fails, there is $\varepsilon > 0$ such that $K^t H_C \not\subset H_C$ for every $t \in (0, \varepsilon)$.

Here $K^t f$ is defined pointwise by $(K^t f)(x) = \mathbb{E}[f(X_t) | X_0 = x]$, which makes sense because every function in H_C is bounded and continuous (Section 3). Under the standing assumptions of [12] this agrees with their definition on $L^2(\mu)$, μ the stationary distribution (Remark 3.1).

Corollary 1.2 (Answer to the question of [12]). *Condition (1.3) implies (1.5), but not conversely. For instance,*

$$A = \begin{pmatrix} -1 & 5 \\ 0 & -1 \end{pmatrix}, \quad B = C = I_2$$

satisfy all standing assumptions of [12] and (1.5), but not (1.3). Hence (1.3) is sufficient but not necessary for $K^t H_C \subset H_C$, $t \geq 0$. Condition (1.5) is exactly condition (1.3) for the matrix $C/\sqrt{2}$ in place of C .

Idea of the proof. Gaussian RKHSs are weighted L^2 -spaces in Fourier variables: a continuous real-valued $f \in L^2(\mathbb{R}^d)$ belongs to H_C if and only if $\int |\widehat{f}(\xi)|^2 e^{\frac{1}{4}\xi^\top C^2 \xi} d\xi < \infty$ (Lemma 2.1). In Fourier variables, K^t acts by composition with the linear map $e^{-A^\top t}$, followed by multiplication with the Gaussian $\exp(-\frac{1}{2}\xi^\top e^{-At}\Sigma(t)e^{-A^\top t}\xi)$ and the constant $e^{-t\text{Tr } A}$ (Lemma 3.2). Invariance therefore amounts to comparing two Gaussian weights, i.e. two quadratic forms. This comparison gives (1.4), and differentiating (1.4) at $t = 0$ gives (1.5).

Where the factor two comes from. The proof in [12] tracks kernel sections. The function $k_z^C = k^C(z, \cdot)$ is a Gaussian bump with “covariance” $\frac{1}{2}C^2$, and averaging over the transition law adds $\Sigma(t)$ to it. This gives $K^t k_z^C = \tau_t k_{e^{-At}z}^{C_t}$ with $C_t := (e^{-At}(C^2 + 2\Sigma(t))e^{-A^\top t})^{1/2}$ and a constant $\tau_t > 0$, and consequently K^t maps H_C into H_{C_t} [12, Proposition 4.1 and its proof]. Moreover, (1.3) is equivalent to $H_{C_t} \subset H_C$ for all $t \geq 0$ (see the proof of [12, Theorem 2.1]). Membership in H_C , however, is decided by $|\widehat{f}|^2$. In $|\widehat{f}|^2$ the Gaussian factor produced by the averaging appears *squared*. As a result, the range of K^t is the space $H_{\tilde{C}_t}$ with $\tilde{C}_t^2 = e^{-At}(C^2 + 4\Sigma(t))e^{-A^\top t}$ (Theorem 4.2), which is strictly smaller than H_{C_t} when $t > 0$ and $B \neq 0$. Invariance is equivalent to $H_{\tilde{C}_t} \subset H_C$, which replaces $2\Sigma(t)$ by $4\Sigma(t)$ and $BB^\top$ by $2BB^\top$. For small t , finite kernel expansions cannot detect non-invariance (Remark 7.2). The witnesses of non-invariance constructed below have Fourier transforms that decay, within a cone of directions, as slowly as membership in H_C permits (Lemma 2.2).

Stable drift. If A is Hurwitz, then $\Sigma := \lim_{t \rightarrow \infty} \Sigma(t)$ exists and $A\Sigma + \Sigma A^\top = -BB^\top$. In this case (1.5) is equivalent to $A(C^2 + 4\Sigma) + (C^2 + 4\Sigma)A^\top \leq 0$, whereas (1.3) is equivalent to $A(C^2 + 2\Sigma) + (C^2 + 2\Sigma)A^\top \leq 0$. Geometrically, H_C is invariant if and only if the ellipsoid $\{x : x^\top(C^2 + 4\Sigma)^{-1}x \leq 1\}$ is forward invariant under $\dot{x} = Ax$ (Theorem 5.1). Moreover, for every C , invariance holds for all sufficiently large t . Figure 1 shows the admissible times for a family of examples.

Beyond Gaussian kernels. The same Fourier argument works for every translation-invariant kernel whose spectral density is positive and integrable (Proposition 7.5). For instance, Sobolev (Matérn) RKHSs are invariant for all A and B (Example 7.6). A Lyapunov-type restriction thus appears because the spectral density of a Gaussian kernel is itself Gaussian.

Organization. Section 2 describes Gaussian RKHSs in Fourier variables, and Section 3 does the same for the Koopman operators. Section 4 proves Theorem 1.1 and Corollary 1.2 and determines the range of K^t . Section 5 treats Hurwitz drifts, Section 6 contains examples, and Section 7 collects further remarks, including other kernels. Appendix A lists numerical checks of the identities.

Notation. For symmetric matrices, $P \leq Q$ means that $Q - P$ is positive semidefinite, and $P > 0$ that P is positive definite. $|x|$ is the Euclidean norm, $\|M\|$ the spectral norm, $M^{-\top} := (M^{-1})^\top$, $\mathbb{S}^{d-1}$ the unit sphere and $\mathbf{1}_E$ the indicator function of a set E . For $x \in \mathbb{R}^d$ we write $k_x^C := k^C(x, \cdot)$.

2. GAUSSIAN RKHSs IN FOURIER VARIABLES

For $g \in L^1(\mathbb{R}^d)$ let $\widehat{g}(\xi) := \int_{\mathbb{R}^d} g(x) e^{-i\xi \cdot x} dx$. The map $g \mapsto \widehat{g}$ extends to $L^2(\mathbb{R}^d)$ with $\|\widehat{g}\|_{L^2}^2 = (2\pi)^d \|g\|_{L^2}^2$ (Plancherel). For $F \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$ put $F^\vee(x) := (2\pi)^{-d} \int F(\xi) e^{i\xi \cdot x} d\xi$. Then $F^\vee$ is continuous, $F^\vee \in L^2(\mathbb{R}^d)$ and $\widehat{F^\vee} = F$. If $g \in L^2(\mathbb{R}^d) \cap C(\mathbb{R}^d)$ and $\widehat{g} \in L^1(\mathbb{R}^d)$, then $g = (\widehat{g})^\vee$ everywhere, since both functions are continuous and agree almost everywhere. A measurable function F is *Hermitian* if $F(-\xi) = \overline{F(\xi)}$ for a.e. ξ . If g is real-valued, then $\widehat{g}$ is Hermitian. If $F \in L^1 \cap L^2$ is Hermitian, then $F^\vee$ is real-valued. For $P > 0$ and $\xi \in \mathbb{R}^d$ we use the Gaussian integral

$$(2.1) \quad \int_{\mathbb{R}^d} \exp(-x^\top P x - i\xi \cdot x) dx = \pi^{d/2} (\det P)^{-1/2} \exp(-\tfrac{1}{4}\xi^\top P^{-1}\xi),$$

which reduces to $\int_{\mathbb{R}} e^{-px^2 - i\xi x} dx = \sqrt{\pi/p} e^{-\xi^2/(4p)}$ after an orthogonal diagonalization of P .

For symmetric positive definite C we define

$$(2.2) \quad w_C(\xi) := \exp(\tfrac{1}{4}\xi^\top C^2 \xi) \quad (\xi \in \mathbb{R}^d), \quad \kappa_C := \pi^{d/2} \det C,$$

and we let $\mathcal{F}_C$ be the set of all measurable Hermitian functions $F: \mathbb{R}^d \rightarrow \mathbb{C}$ with $\int |F|^2 w_C d\xi < \infty$. By (2.1) with $P = C^{-2}$, followed by a translation,

$$(2.3) \quad \widehat{k_y^C}(\xi) = \kappa_C e^{-i\xi \cdot y} w_C(\xi)^{-1}, \quad y, \xi \in \mathbb{R}^d,$$

and by (2.1) with $\xi = 0$ and $P = \tfrac{1}{4}C^2$,

$$(2.4) \quad \int_{\mathbb{R}^d} w_C(\xi)^{-1} d\xi = \frac{(4\pi)^{d/2}}{\det C} < \infty.$$

Lemma 2.1 (Fourier description of H_C). *Let $C \in \mathbb{R}^{d \times d}$ be symmetric positive definite.*

(a) *Every $F \in \mathcal{F}_C$ belongs to $L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$, and*

$$\|F\|_{L^1} \leq \left(\frac{(4\pi)^{d/2}}{\det C} \right)^{1/2} \left(\int |F|^2 w_C d\xi \right)^{1/2}.$$

(b) $H_C = \{f \in C(\mathbb{R}^d) \cap L^2(\mathbb{R}^d) \text{ real-valued} : \widehat{f} \in \mathcal{F}_C\}$, and for $f, g \in H_C$

$$(2.5) \quad \langle f, g \rangle_C = (2\pi)^{-d} \kappa_C^{-1} \int_{\mathbb{R}^d} \widehat{f}(\xi) \overline{\widehat{g}(\xi)} w_C(\xi) d\xi.$$

(c) *The map $f \mapsto \widehat{f}$ is a bijection of H_C onto $\mathcal{F}_C$, with inverse $F \mapsto F^\vee$.*

Proof. (a) By the Cauchy–Schwarz inequality and (2.4),

$$\int |F| d\xi = \int |F| w_C^{1/2} w_C^{-1/2} d\xi \leq \left(\int |F|^2 w_C d\xi \right)^{1/2} \left(\int w_C^{-1} d\xi \right)^{1/2}.$$

Since $w_C \geq 1$, also $\int |F|^2 d\xi \leq \int |F|^2 w_C d\xi$.

(b) Let $\mathcal{H}$ be the set on the right-hand side, equipped with the form $\langle f, g \rangle_{\mathcal{H}}$ defined by the right-hand side of (2.5).

$\mathcal{H}$ is a real inner product space. For real f, g the functions $\widehat{f}, \widehat{g}$ are Hermitian and w_C is even. So the integrand at $-\xi$ is the complex conjugate of the integrand at ξ , and the integral is real. The form is bilinear, symmetric and nonnegative. If $\langle f, f \rangle_{\mathcal{H}} = 0$, then $\widehat{f} = 0$ a.e., hence $f = 0$ in L^2 , and hence $f = 0$ by continuity.

Inversion. For $f \in \mathcal{H}$ we have $\widehat{f} \in L^1$ by (a), and therefore

$$(2.6) \quad f(x) = (2\pi)^{-d} \int \widehat{f}(\xi) e^{i\xi \cdot x} d\xi \quad \text{for every } x \in \mathbb{R}^d.$$

Completeness. Let (f_n) be a Cauchy sequence in $\mathcal{H}$. Then $(\widehat{f_n})$ is Cauchy in $L^2(w_C d\xi)$ and converges there to some F . A subsequence converges a.e., so F is Hermitian, and $F \in \mathcal{F}_C$. By (a), $F \in L^1 \cap L^2$. The function $f := F^\vee$ is continuous, real-valued and in L^2 , with $\widehat{f} = F$. Hence $f \in \mathcal{H}$ and $\|f_n - f\|_{\mathcal{H}}^2 = (2\pi)^{-d} \kappa_C^{-1} \int |\widehat{f_n} - F|^2 w_C \rightarrow 0$.

Kernel sections. By (2.3) and (2.4), $\widehat{k_y^C}$ is Hermitian and $\int |\widehat{k_y^C}|^2 w_C = \kappa_C^2 \int w_C^{-1} < \infty$. Since k_y^C is real, continuous and in L^2 , it belongs to $\mathcal{H}$.

Reproducing property. For $f \in \mathcal{H}$ and $y \in \mathbb{R}^d$, (2.3) and (2.6) give

$$\begin{aligned} \langle f, k_y^C \rangle_{\mathcal{H}} &= (2\pi)^{-d} \kappa_C^{-1} \int \widehat{f}(\xi) \kappa_C e^{i\xi \cdot y} w_C(\xi)^{-1} w_C(\xi) d\xi \\ &= (2\pi)^{-d} \int \widehat{f}(\xi) e^{i\xi \cdot y} d\xi = f(y). \end{aligned}$$

Identification with H_C . This is the uniqueness part of the Moore–Aronszajn theorem [1, 10]; we include the short argument. Let $H_0 := \text{span}\{k_y^C : y \in \mathbb{R}^d\}$. If $f \perp H_0$ in $\mathcal{H}$, then $f(y) = \langle f, k_y^C \rangle_{\mathcal{H}} = 0$ for all y . Hence H_0 is dense in $\mathcal{H}$, and by the same argument it is dense in H_C . Both norms coincide on H_0 , since $\|\sum_j \alpha_j k_{y_j}^C\|^2 = \sum_{i,j} \alpha_i \alpha_j k^C(y_i, y_j)$ in either space. In both spaces, norm convergence implies pointwise convergence, because $|h(x)| \leq \|h\| k^C(x, x)^{1/2} = \|h\|$.

Now let $f \in \mathcal{H}$ and choose $f_n \in H_0$ with $f_n \rightarrow f$ in $\mathcal{H}$. Then (f_n) is Cauchy in H_C , so $f_n \rightarrow g$ in H_C for some g , and comparing pointwise limits gives $g = f$. Thus $f \in H_C$ and $\|f\|_C = \lim \|f_n\|_C = \lim \|f_n\|_{\mathcal{H}} = \|f\|_{\mathcal{H}}$. Exchanging the roles of $\mathcal{H}$ and H_C gives $H_C \subset \mathcal{H}$. Hence $\mathcal{H} = H_C$ isometrically, and (2.5) follows by polarization.

(c) By (b), $\widehat{f} \in \mathcal{F}_C$ for $f \in H_C$. The map is injective because $\widehat{f}$ determines the continuous L^2 -function f . It is surjective: for $F \in \mathcal{F}_C$, part (a) gives

$F \in L^1 \cap L^2$, so $F^\vee$ is real, continuous and in L^2 with $\widehat{F^\vee} = F$, and $F^\vee \in H_C$ by (b). $\square$

Lemma 2.1 is the Gaussian case of the Fourier description of native spaces of translation-invariant kernels [16, Theorem 10.12]. A power-series description of Gaussian RKHSs is given in [13]. The next lemma is the tool for proving that an inclusion fails.

Lemma 2.2 (Exponential multipliers). *Let C be symmetric positive definite and $S \in \mathbb{R}^{d \times d}$ symmetric. Then*

$$\int_{\mathbb{R}^d} |F(\xi)|^2 w_C(\xi) \exp\left(\frac{1}{4}\xi^\top S \xi\right) d\xi < \infty \quad \text{for every } F \in \mathcal{F}_C$$

if and only if $S \leq 0$. If $S \leq 0$ fails, there is a real-valued even function $F_S \in \mathcal{F}_C$, supported in an open double cone, for which the integral is $+\infty$.

Proof. If $S \leq 0$, then $\exp(\frac{1}{4}\xi^\top S \xi) \leq 1$. Conversely, suppose that $S \leq 0$ fails, and let $\lambda > 0$ be an eigenvalue of S with unit eigenvector v . The set $U := \{\theta \in \mathbb{S}^{d-1} : \theta^\top S \theta > \lambda/2\}$ is relatively open in $\mathbb{S}^{d-1}$, satisfies $U = -U$ and contains v . Hence $\Gamma := \{r\theta : r > 0, \theta \in U\}$ is an open double cone, and $\xi^\top S \xi > \frac{\lambda}{2}|\xi|^2$ for $\xi \in \Gamma$. Define

$$F_S(\xi) := w_C(\xi)^{-1/2} (1 + |\xi|^2)^{-d/2} \mathbf{1}_\Gamma(\xi).$$

Since $\Gamma = -\Gamma$ and w_C is even, F_S is real and even, hence Hermitian. Moreover $\int |F_S|^2 w_C = \int_\Gamma (1 + |\xi|^2)^{-d} d\xi \leq \int_{\mathbb{R}^d} (1 + |\xi|^2)^{-d} d\xi < \infty$, because the radial integrand $r^{d-1}(1+r^2)^{-d}$ is $O(r^{-d-1})$ as $r \rightarrow \infty$. So $F_S \in \mathcal{F}_C$. In polar coordinates, with $|U| > 0$ the surface measure of U (the counting measure if $d = 1$),

$$\begin{aligned} \int |F_S|^2 w_C e^{\frac{1}{4}\xi^\top S \xi} d\xi &\geq \int_\Gamma (1 + |\xi|^2)^{-d} e^{\lambda|\xi|^2/8} d\xi \\ &= |U| \int_0^\infty r^{d-1} (1 + r^2)^{-d} e^{\lambda r^2/8} dr = +\infty. \quad \square \end{aligned}$$

As a first application we obtain a short proof of the inclusion criterion [12, Proposition 3.3].

Corollary 2.3. *For symmetric positive definite C_1, C_2 we have $H_{C_1} \subset H_{C_2}$ if and only if $C_1^2 \geq C_2^2$. In this case $\|f\|_{C_2} \leq (\det C_1 / \det C_2)^{1/2} \|f\|_{C_1}$ for $f \in H_{C_1}$.*

Proof. By Lemma 2.1(b),(c), $H_{C_1} \subset H_{C_2}$ holds if and only if $\int |F|^2 w_{C_2} < \infty$ for every $F \in \mathcal{F}_{C_1}$. Since $w_{C_2} = w_{C_1} \exp(\frac{1}{4}\xi^\top (C_2^2 - C_1^2)\xi)$, Lemma 2.2 with $S = C_2^2 - C_1^2$ gives the equivalence. If $C_1^2 \geq C_2^2$, then $w_{C_2} \leq w_{C_1}$, and (2.5) yields $\|f\|_{C_2}^2 \leq (\kappa_{C_1}/\kappa_{C_2}) \|f\|_{C_1}^2 = (\det C_1 / \det C_2) \|f\|_{C_1}^2$. $\square$

3. THE KOOPMAN OPERATORS IN FOURIER VARIABLES

Let $\gamma_t := \mathcal{N}(0, \Sigma(t))$, a centered Gaussian measure on $\mathbb{R}^d$ that may be degenerate ($\gamma_0 = \delta_0$). The solution of (1.1) with $X_0 = x$ is $X_t = e^{At}x + Z_t$, where $Z_t = \int_0^t e^{A(t-s)}B \, dW_s$ is centered Gaussian with covariance $\int_0^t e^{A(t-s)}BB^\top e^{A^\top(t-s)} \, ds = \Sigma(t)$ by the Itô isometry (see also [15]). Hence, for bounded Borel functions f ,

$$(3.1) \quad (K^t f)(x) = \int_{\mathbb{R}^d} f(e^{At}x + y) \gamma_t(dy), \quad x \in \mathbb{R}^d.$$

Every $f \in H_C$ is continuous by Lemma 2.1(b), and bounded, since $|f(x)| = |\langle f, k_x^C \rangle_C| \leq \|f\|_C \|k^C(x, x)\|^{1/2} = \|f\|_C$.

Remark 3.1 (Two readings of $K^t H_C \subset H_C$). Under the standing assumptions of [12], K^t is considered as an operator on $L^2(\mu)$, where $\mu = \mathcal{N}(0, \Sigma)$ is the stationary distribution, and H_C is identified with a subspace of $L^2(\mu)$ [12, Lemma 3.1]. By Lemma 3.2 below, $K^t f$ is continuous for $f \in H_C$. Since μ has a positive density and the elements of H_C are continuous, the $L^2(\mu)$ -class of $K^t f$ contains an element of H_C if and only if the continuous function $K^t f$ belongs to H_C . So the two readings of “ $K^t H_C \subset H_C$ ” agree.

Lemma 3.2. *Let f be bounded, continuous and in $L^2(\mathbb{R}^d)$, and let $t \geq 0$. Then $K^t f$ is bounded, continuous and in $L^2(\mathbb{R}^d)$, and for a.e. ξ*

$$(3.2) \quad \widehat{K^t f}(\xi) = e^{-t \operatorname{Tr} A} \widehat{f}(e^{-A^\top t} \xi) \exp\left(-\frac{1}{2} \xi^\top e^{-At} \Sigma(t) e^{-A^\top t} \xi\right).$$

Proof. Boundedness is clear, and continuity follows from (3.1) by dominated convergence. For $g \in L^2(\mathbb{R}^d)$ put $(J_t g)(z) := \int g(z + y) \gamma_t(dy)$. Minkowski’s integral inequality, applied to $|g|$, shows that the integral converges absolutely for a.e. z and that $\|J_t g\|_{L^2} \leq \int \|g(\cdot + y)\|_{L^2} \gamma_t(dy) = \|g\|_{L^2}$. For $g \in L^1 \cap L^2$, Fubini’s theorem and the characteristic function of γ_t give

$$\widehat{J_t g}(\xi) = \int \int g(z + y) e^{-i\xi \cdot z} \, dz \, \gamma_t(dy) = \widehat{g}(\xi) \int e^{i\xi \cdot y} \gamma_t(dy) = \widehat{g}(\xi) e^{-\frac{1}{2} \xi^\top \Sigma(t) \xi}.$$

Both sides are bounded linear maps from L^2 to L^2 in g , so the identity extends from the dense subspace $L^1 \cap L^2$ to all of L^2 . For bounded f the integral defining $J_t f$ converges everywhere, and (3.1) says that $K^t f = \widehat{(J_t f) \circ e^{At}}$. Finally, for $u \in L^2$ and invertible E we have $u \circ E \in L^2$ and $\widehat{u \circ E}(\xi) = |\det E|^{-1} \widehat{u}(E^{-\top} \xi)$; this follows by the substitution $x = E^{-1}z$ for $u \in L^1 \cap L^2$ and extends by density. With $E = e^{At}$, $\det E = e^{t \operatorname{Tr} A}$ and $E^{-\top} = e^{-A^\top t}$, this yields (3.2), because $(e^{-A^\top t} \xi)^\top \Sigma(t) (e^{-A^\top t} \xi) = \xi^\top e^{-At} \Sigma(t) e^{-A^\top t} \xi$. $\square$

We introduce the symmetric matrices

$$(3.3) \quad D_t := C^2 + 4\Sigma(t) - e^{At} C^2 e^{A^\top t} \quad (t \geq 0), \quad \Lambda_C := 4BB^\top - (AC^2 + C^2 A^\top).$$

Thus (1.4) reads $D_t \geq 0$, (1.5) reads $\Lambda_C \geq 0$, and (1.3) reads $2BB^\top - (AC^2 + C^2 A^\top) \geq 0$.

Proposition 3.3 (Norm identity). *For $f \in H_C$ and $t \geq 0$ the following identity holds in $[0, \infty]$:*

$$(3.4) \quad \int_{\mathbb{R}^d} |\widehat{K^t f}(\xi)|^2 w_C(\xi) d\xi = e^{-t \operatorname{Tr} A} \int_{\mathbb{R}^d} |\widehat{f}(\eta)|^2 w_C(\eta) e^{-\frac{1}{4} \eta^\top D_t \eta} d\eta.$$

Proof. By (3.2), the left-hand side equals

$$e^{-2t \operatorname{Tr} A} \int |\widehat{f}(e^{-A^\top t} \xi)|^2 \exp(-\xi^\top e^{-At} \Sigma(t) e^{-A^\top t} \xi + \frac{1}{4} \xi^\top C^2 \xi) d\xi.$$

The substitution $\xi = e^{A^\top t} \eta$, $d\xi = e^{t \operatorname{Tr} A} d\eta$, turns this into

$$e^{-t \operatorname{Tr} A} \int |\widehat{f}(\eta)|^2 \exp\left(-\eta^\top \Sigma(t) \eta + \frac{1}{4} \eta^\top e^{At} C^2 e^{A^\top t} \eta\right) d\eta,$$

and $-\eta^\top \Sigma(t) \eta + \frac{1}{4} \eta^\top e^{At} C^2 e^{A^\top t} \eta = \frac{1}{4} \eta^\top C^2 \eta - \frac{1}{4} \eta^\top D_t \eta$. $\square$

4. PROOFS OF THEOREM 1.1 AND COROLLARY 1.2

Throughout this section $A \in \mathbb{R}^{d \times d}$ and $B \in \mathbb{R}^{d \times m}$ are arbitrary and C is symmetric positive definite.

Theorem 4.1 (Fixed time). *Let $t \geq 0$.*

- (a) $K^t H_C \subset H_C$ if and only if $D_t \geq 0$.
- (b) If $D_t \geq 0$, then K^t is bounded on H_C and $\|K^t\|_{H_C \rightarrow H_C} = e^{-t \operatorname{Tr} A/2}$.
- (c) If $D_t \geq 0$ fails, there is $f \in H_C$ with $K^t f \notin H_C$; one can choose f such that $\widehat{f}$ is real, even and supported in an open double cone.

Proof. Let $f \in H_C$. By Lemma 3.2, $K^t f$ is real, continuous and in L^2 . By Lemma 2.1(b), $K^t f \in H_C$ if and only if the left-hand side of (3.4) is finite, and then $\|K^t f\|_C^2$ equals $(2\pi)^{-d} \kappa_C^{-1}$ times that quantity.

Sufficiency in (a) and the upper bound in (b). If $D_t \geq 0$, then the factor $e^{-\frac{1}{4} \eta^\top D_t \eta}$ in (3.4) is at most 1. Hence $K^t f \in H_C$ and, by (2.5), $\|K^t f\|_C^2 \leq e^{-t \operatorname{Tr} A} \|f\|_C^2$.

Lower bound in (b). For $\varepsilon > 0$ the indicator $F_\varepsilon := \mathbf{1}_{\{|\eta| < \varepsilon\}}$ is real, even and belongs to $\mathcal{F}_C$, so $f_\varepsilon := F_\varepsilon^\vee \in H_C \setminus \{0\}$ by Lemma 2.1(c). Since $\eta^\top D_t \eta \leq \|D_t\| \varepsilon^2$ for $|\eta| < \varepsilon$, (3.4) and (2.5) give

$$\begin{aligned} \|K^t f_\varepsilon\|_C^2 &= e^{-t \operatorname{Tr} A} (2\pi)^{-d} \kappa_C^{-1} \int_{|\eta| < \varepsilon} w_C(\eta) e^{-\frac{1}{4} \eta^\top D_t \eta} d\eta \\ &\geq e^{-t \operatorname{Tr} A} e^{-\frac{1}{4} \varepsilon^2 \|D_t\|} \|f_\varepsilon\|_C^2. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ gives $\|K^t\|^2 \geq e^{-t \operatorname{Tr} A}$.

Necessity in (a), and (c). Suppose that $D_t \geq 0$ fails. Lemma 2.2 with $S := -D_t$ yields a real, even $F_S \in \mathcal{F}_C$, supported in an open double cone, with $\int |F_S|^2 w_C e^{-\frac{1}{4} \eta^\top D_t \eta} d\eta = \infty$. Let $f := F_S^\vee$. Then $f \in H_C$ and $\widehat{f} = F_S$ by Lemma 2.1(c), and by (3.4) $\int |\widehat{K^t f}|^2 w_C d\xi = \infty$. Hence $K^t f \notin H_C$. $\square$

The next result identifies the range of K^t . It shows exactly what is lost in the approach of [12].

Theorem 4.2 (Range of K^t). *For $t \geq 0$ let*

$$(4.1) \quad \tilde{C}_t := \left(e^{-At} (C^2 + 4\Sigma(t)) e^{-A^\top t} \right)^{1/2}, \quad \tilde{\tau}_t := \frac{\det C}{(\det(C^2 + 4\Sigma(t)))^{1/2}}.$$

Then K^t maps H_C bijectively onto $H_{\tilde{C}_t}$, and $\|K^t f\|_{\tilde{C}_t}^2 = \tilde{\tau}_t \|f\|_C^2$ for all $f \in H_C$.

Proof. Let $f \in H_C$ and $g := K^t f$, which is real, continuous and in L^2 by Lemma 3.2. By (3.2) and the substitution $\xi = e^{A^\top t} \eta$, exactly as in the proof of Proposition 3.3,

$$\begin{aligned} \int |\hat{g}(\xi)|^2 w_{\tilde{C}_t}(\xi) d\xi &= e^{-t \operatorname{Tr} A} \int |\hat{f}(\eta)|^2 e^{-\eta^\top \Sigma(t) \eta + \frac{1}{4} \eta^\top e^{At} \tilde{C}_t^2 e^{A^\top t} \eta} d\eta \\ &= e^{-t \operatorname{Tr} A} \int |\hat{f}(\eta)|^2 w_C(\eta) d\eta, \end{aligned}$$

because $e^{At} \tilde{C}_t^2 e^{A^\top t} = C^2 + 4\Sigma(t)$. Thus $g \in H_{\tilde{C}_t}$ by Lemma 2.1(b), applied with $\tilde{C}_t$ in place of C . Since $\det \tilde{C}_t = e^{-t \operatorname{Tr} A} \det(C^2 + 4\Sigma(t))^{1/2}$, (2.5) gives

$$\|g\|_{\tilde{C}_t}^2 = \frac{\kappa_C}{\kappa_{\tilde{C}_t}} e^{-t \operatorname{Tr} A} \|f\|_C^2 = \frac{\det C}{\det \tilde{C}_t} e^{-t \operatorname{Tr} A} \|f\|_C^2 = \tilde{\tau}_t \|f\|_C^2.$$

In particular, K^t is injective on H_C .

For surjectivity, let $g \in H_{\tilde{C}_t}$ and define

$$F(\eta) := e^{t \operatorname{Tr} A} \hat{g}(e^{A^\top t} \eta) \exp\left(\frac{1}{2} \eta^\top \Sigma(t) \eta\right), \quad \eta \in \mathbb{R}^d.$$

Then F is Hermitian. With $\eta = e^{-A^\top t} \xi$, $d\eta = e^{-t \operatorname{Tr} A} d\xi$ and $\eta^\top \Sigma(t) \eta + \frac{1}{4} \eta^\top C^2 \eta = \frac{1}{4} \xi^\top \tilde{C}_t^2 \xi$ we get

$$\begin{aligned} \int |F(\eta)|^2 w_C(\eta) d\eta &= e^{2t \operatorname{Tr} A} \int |\hat{g}(e^{A^\top t} \eta)|^2 e^{\eta^\top \Sigma(t) \eta + \frac{1}{4} \eta^\top C^2 \eta} d\eta \\ &= e^{t \operatorname{Tr} A} \int |\hat{g}(\xi)|^2 w_{\tilde{C}_t}(\xi) d\xi < \infty. \end{aligned}$$

Hence $F \in \mathcal{F}_C$, and $f := F^\vee \in H_C$ by Lemma 2.1(c). By (3.2),

$$\widehat{K^t f}(\xi) = e^{-t \operatorname{Tr} A} F(e^{-A^\top t} \xi) \exp\left(-\frac{1}{2} \xi^\top e^{-At} \Sigma(t) e^{-A^\top t} \xi\right) = \hat{g}(\xi).$$

Since $K^t f$ and g are continuous L^2 -functions with the same Fourier transform, $K^t f = g$. $\square$

Corollary 4.3 (On [12, Remark 4.2]). *For every $t \geq 0$,*

$$K^t H_C \subset H_C \iff H_{\tilde{C}_t} \subset H_C \iff \tilde{C}_t^2 \geq C^2 \iff D_t \geq 0.$$

Let $C_t := (e^{-At} (C^2 + 2\Sigma(t)) e^{-A^\top t})^{1/2}$ be the matrix of [12, (4.1)]. Then $\tilde{C}_t^2 - C_t^2 = 2e^{-At} \Sigma(t) e^{-A^\top t} \geq 0$, so $H_{\tilde{C}_t} \subset H_{C_t}$, and the inclusion is strict whenever $\Sigma(t) \neq 0$, in particular for every $t > 0$ if $B \neq 0$. The implication “ $K^t H_C \subset H_C \Rightarrow H_{C_t} \subset H_C$ ” discussed in [12, Remark 4.2] is false in

general. For A, B, C as in Corollary 1.2, $K^t H_C \subset H_C$ for every $t \geq 0$, whereas there is $\varepsilon > 0$ with $H_{C_t} \not\subset H_C$ for all $t \in (0, \varepsilon)$.

Proof. The first equivalence is Theorem 4.2, the second is Corollary 2.3, and the third follows from the congruence $\tilde{C}_t^2 - C^2 = e^{-At} D_t e^{-A^\top t}$. The formula for $\tilde{C}_t^2 - C_t^2$ is immediate, and Corollary 2.3 gives $H_{\tilde{C}_t} \subset H_{C_t}$. If also $H_{C_t} \subset H_{\tilde{C}_t}$, then $C_t^2 \geq \tilde{C}_t^2$, i.e. $-2e^{-At} \Sigma(t) e^{-A^\top t} \geq 0$, which forces $\Sigma(t) = 0$. If $B \neq 0$ and $t > 0$, then $\Sigma(t) \neq 0$, since the continuous positive semidefinite integrand in (1.2) equals $BB^\top \neq 0$ at $s = 0$.

For the last assertion, let $f_x(t) := \langle C_t^2 x, x \rangle$ as in the proof of [12, Theorem 2.1]. Since $\Sigma(0) = 0$ and $\dot{\Sigma}(0) = BB^\top$, differentiating $C_t^2 = e^{-At}(C^2 + 2\Sigma(t))e^{-A^\top t}$ at $t = 0$ gives $\dot{f}_x(0) = \langle (2BB^\top - (AC^2 + C^2A^\top))x, x \rangle$. For the matrices of Corollary 1.2, (1.3) fails, so $\dot{f}_x(0) < 0$ for some x . Then $f_x(t) < f_x(0) = |Cx|^2$ for $t \in (0, \varepsilon)$ and some $\varepsilon > 0$, i.e. $C_t^2 \geq C^2$ fails, and Corollary 2.3 gives $H_{C_t} \not\subset H_C$. Invariance for all t follows from Theorem 4.4 below, since (1.5) holds for these matrices. $\square$

Theorem 4.4 (The whole semigroup). *The following statements are equivalent:*

- (i) $K^t H_C \subset H_C$ for every $t \geq 0$;
- (ii) there is a sequence $t_n \downarrow 0$ with $t_n > 0$ and $K^{t_n} H_C \subset H_C$ for every n ;
- (iii) $\frac{1}{2}(AC^2 + C^2A^\top) \leq 2BB^\top$, i.e. $\Lambda_C \geq 0$.

Proof. We have $D_0 = 0$ and, since A commutes with e^{At} ,

$$\frac{d}{dt} D_t = 4e^{At} BB^\top e^{A^\top t} - Ae^{At} C^2 e^{A^\top t} - e^{At} C^2 e^{A^\top t} A^\top = e^{At} \Lambda_C e^{A^\top t}.$$

Hence

$$(4.2) \quad D_t = \int_0^t e^{As} \Lambda_C e^{A^\top s} ds \quad (t \geq 0).$$

(iii) $\Rightarrow$ (i): If $\Lambda_C \geq 0$, then each matrix $e^{As} \Lambda_C e^{A^\top s}$ is positive semidefinite, so $D_t \geq 0$ for all t , and Theorem 4.1(a) applies. (i) $\Rightarrow$ (ii) is trivial. (ii) $\Rightarrow$ (iii): By Theorem 4.1(a), $t_n^{-1} D_{t_n} \geq 0$ for all n . By (4.2) and the continuity of the integrand, $t^{-1} D_t \rightarrow \Lambda_C$ as $t \downarrow 0$. Since the cone of positive semidefinite matrices is closed, $\Lambda_C \geq 0$. $\square$

Remark 4.5. Theorem 4.4 is the computation in the proof of [12, Theorem 2.1], with C_t replaced by $\tilde{C}_t$. Indeed, $\tilde{f}_x(t) := \langle \tilde{C}_t^2 x, x \rangle = |C e^{-A^\top t} x|^2 + 4 \int_0^t |B^\top e^{-A^\top s} x|^2 ds$ has derivative $\dot{\tilde{f}}_x(t) = \langle \Lambda_C e^{-A^\top t} x, e^{-A^\top t} x \rangle$. Replacing C_t by $\tilde{C}_t$ is what turns the sufficient condition into a characterization.

Proof of Theorem 1.1. Part (a) is Theorem 4.1(a),(b), since $D_t \geq 0$ is (1.4). The equivalence in part (b) is (i) $\Leftrightarrow$ (iii) of Theorem 4.4. If (1.5) fails, then (ii) of Theorem 4.4 fails. That is, no sequence $t_n \downarrow 0$ with $K^{t_n} H_C \subset H_C$ exists, which means that $K^t H_C \not\subset H_C$ for all t in some interval $(0, \varepsilon)$. $\square$

Proof of Corollary 1.2. Since $BB^\top \geq 0$, we have $BB^\top \leq 2BB^\top$, so (1.3) implies (1.5). For the matrices in Corollary 1.2, A is Hurwitz (double eigenvalue -1), (A, B) is controllable because B is invertible, and $m = d = 2$. Moreover $\frac{1}{2}(AC^2 + C^2A^\top) = \frac{1}{2}(A + A^\top)$ has eigenvalues $\frac{1}{2}(-2 \pm 5)$, the largest being $\frac{3}{2}$. Hence (1.3) ($\frac{3}{2} \leq 1$) fails and (1.5) ($\frac{3}{2} \leq 2$) holds. By Theorem 1.1(b), H_C is invariant under every K^t . The last statement follows from $\frac{1}{2}(A(C/\sqrt{2})^2 + (C/\sqrt{2})^2A^\top) = \frac{1}{4}(AC^2 + C^2A^\top)$. $\square$

For isotropic kernels, $C = \sigma I$, Theorem 1.1 sharpens [12, Corollary 2.3] as follows.

Corollary 4.6. *Let $\sigma > 0$. The RKHS $H_\sigma := H_{\sigma I}$ of $\exp(-|x - y|^2/\sigma^2)$ is invariant under every K^t , $t \geq 0$, if and only if $\frac{1}{2}(A + A^\top) \leq \frac{2}{\sigma^2}BB^\top$. In that case $\|K^t\|_{H_\sigma \rightarrow H_\sigma} = e^{-t \operatorname{Tr} A/2}$.*

Proof. For $C = \sigma I$, condition (1.5) reads $\frac{\sigma^2}{2}(A + A^\top) \leq 2BB^\top$. Apply Theorem 1.1. $\square$

5. STABLE DRIFT: INVARIANT ELLIPSOIDS AND ADMISSIBLE TIMES

Theorem 5.1. *Let A be Hurwitz (controllability is not needed), and let $\Sigma := \int_0^\infty e^{As}BB^\top e^{A^\top s} ds$. Put $P_C := C^2 + 4\Sigma$, which is positive definite, and let $\mathcal{E}_C := \{x \in \mathbb{R}^d : |x|_{P_C} \leq 1\}$ be the unit ball of the norm $|x|_{P_C} := (x^\top P_C^{-1}x)^{1/2}$.*

(a) $D_t = P_C - e^{At}P_Ce^{A^\top t}$ for all $t \geq 0$. Consequently, for each $t \geq 0$,

$$\begin{aligned} K^t H_C \subset H_C &\iff e^{At}P_Ce^{A^\top t} \leq P_C \\ &\iff |e^{At}x|_{P_C} \leq |x|_{P_C} \text{ for all } x \in \mathbb{R}^d \\ &\iff e^{At}\mathcal{E}_C \subset \mathcal{E}_C. \end{aligned}$$

(b) $AP_C + P_CA^\top = -\Lambda_C$. Hence (1.5) holds if and only if $AP_C + P_CA^\top \leq 0$, i.e. if and only if $\mathcal{E}_C$ is forward invariant under $\dot{x} = Ax$. Likewise, (1.3) holds if and only if $A(C^2 + 2\Sigma) + (C^2 + 2\Sigma)A^\top \leq 0$.

(c) The set $T_C := \{t \geq 0 : K^t H_C \subset H_C\}$ is a closed additive sub-semigroup of $[0, \infty)$. It contains 0, and it contains $[t_C, \infty)$ for every $t_C \geq 0$ with $\sup_{t \geq t_C} \|e^{At}\|^2 \leq \lambda_{\min}(P_C)/\lambda_{\max}(P_C)$; such t_C exist. If (1.5) holds, then $T_C = [0, \infty)$; otherwise $T_C \cap (0, \varepsilon) = \emptyset$ for some $\varepsilon > 0$.

Proof. (a) Substituting $s = t + r$ in $\int_t^\infty e^{As}BB^\top e^{A^\top s} ds$ gives $\Sigma(t) = \Sigma - e^{At}\Sigma e^{A^\top t}$. Inserting this into (3.3) yields $D_t = C^2 + 4\Sigma - e^{At}(C^2 + 4\Sigma)e^{A^\top t} = P_C - e^{At}P_Ce^{A^\top t}$, so the first equivalence is Theorem 4.1(a). Write $P = P_C$, $E = e^{At}$ and $G := P^{-1/2}EP^{1/2}$. Since $P^{-1/2}EPE^\top P^{-1/2} = GG^\top$, we have $EPE^\top \leq P \iff GG^\top \leq I \iff \|G^\top\| \leq 1 \iff \|G\| \leq 1$. Moreover $\|G\| \leq 1$ means $|P^{-1/2}EP^{1/2}z| \leq |z|$ for all z , i.e. (with $x = P^{1/2}z$)

$|Ex|_P \leq |x|_P$ for all x . Since $|\cdot|_P$ is a norm with unit ball $\mathcal{E}_C$, this is equivalent to $E\mathcal{E}_C \subset \mathcal{E}_C$.

(b) Since A is Hurwitz, $A\Sigma + \Sigma A^\top = \int_0^\infty \frac{d}{ds}(e^{As}BB^\top e^{A^\top s}) ds = -BB^\top$. Hence $AP_C + P_CA^\top = AC^2 + C^2A^\top - 4BB^\top = -\Lambda_C$ and $A(C^2 + 2\Sigma) + (C^2 + 2\Sigma)A^\top = AC^2 + C^2A^\top - 2BB^\top$. By (a), forward invariance of $\mathcal{E}_C$ means $e^{At}P_Ce^{A^\top t} \leq P_C$ for all $t \geq 0$. If $AP_C + P_CA^\top \leq 0$, this follows from $\frac{d}{dt}e^{At}P_Ce^{A^\top t} = e^{At}(AP_C + P_CA^\top)e^{A^\top t} \leq 0$. Conversely, dividing $P_C - e^{At}P_Ce^{A^\top t} \geq 0$ by t and letting $t \downarrow 0$ gives $-(AP_C + P_CA^\top) \geq 0$.

(c) We have $0 \in T_C$ because $D_0 = 0$, and T_C is closed because $t \mapsto D_t$ is continuous and the positive semidefinite cone is closed. If $s, t \in T_C$, then, since congruence preserves the Loewner order, $e^{A(s+t)}P_Ce^{A^\top(s+t)} = e^{As}(e^{At}P_Ce^{A^\top t})e^{A^\top s} \leq e^{As}P_Ce^{A^\top s} \leq P_C$, so $s+t \in T_C$. For $t \geq t_C$ we have $e^{At}P_Ce^{A^\top t} \leq \lambda_{\max}(P_C)\|e^{At}\|^2I \leq \lambda_{\min}(P_C)I \leq P_C$. Such t_C exist because $\|e^{At}\| \rightarrow 0$ as $t \rightarrow \infty$. The final dichotomy is Theorem 1.1(b). $\square$

Remark 5.2. Since $A\Sigma + \Sigma A^\top = -BB^\top \leq 0$, the ellipsoid of Σ is forward invariant whenever $\Sigma > 0$ (e.g. if (A, B) is controllable). By Theorem 5.1, H_C is Koopman invariant precisely when the ellipsoid of $\Sigma + \frac{1}{4}C^2$ is still forward invariant; condition (1.3) requires this for $\Sigma + \frac{1}{2}C^2$. Theorem 5.1(c) also shows that, for Hurwitz A , every Gaussian RKHS is invariant under K^t for all sufficiently large t , even when invariance fails for all small $t > 0$ (Example 6.1, $\beta > 6$).

6. EXAMPLES

Example 6.1 (A one-parameter family). Let $d = m = 2$, $B = C = I_2$, and

$$A = A_\beta := \begin{pmatrix} -1 & \beta \\ 0 & -1 \end{pmatrix}, \quad \beta \geq 0.$$

Then A_β is Hurwitz (double eigenvalue -1), and (A_β, B) is controllable since B is invertible. The matrix $AC^2 + C^2A^\top = A_\beta + A_\beta^\top$ has eigenvalues $-2 \pm \beta$, so

$$(1.3) \iff \frac{1}{2}(\beta - 2) \leq 1 \iff \beta \leq 4,$$

$$(1.5) \iff \frac{1}{2}(\beta - 2) \leq 2 \iff \beta \leq 6.$$

For $4 < \beta \leq 6$, H_I is therefore invariant under every K^t , with $\|K^t\|_{H_I \rightarrow H_I} = e^t$ (as $\text{Tr } A_\beta = -2$), although (1.3) fails. For $\beta > 6$ there is $\varepsilon > 0$ with $K^t H_I \not\subset H_I$ for all $t \in (0, \varepsilon)$.

Everything here is explicit. We have $A_\beta = -I + \beta E_{12}$, where E_{12} has the entry 1 in position $(1, 2)$ and zeros elsewhere. Since $E_{12}^2 = 0$,

$$e^{A_\beta t} = e^{-t} \begin{pmatrix} 1 & \beta t \\ 0 & 1 \end{pmatrix}, \quad \Sigma(t) = \int_0^t e^{-2s} \begin{pmatrix} 1 + \beta^2 s^2 & \beta s \\ \beta s & 1 \end{pmatrix} ds.$$

The elementary integrals

$$\begin{aligned} \int_0^t e^{-2s} ds &= \frac{1 - e^{-2t}}{2}, & \int_0^t s e^{-2s} ds &= \frac{1 - e^{-2t}(1 + 2t)}{4}, \\ \int_0^t s^2 e^{-2s} ds &= \frac{1 - e^{-2t}(1 + 2t + 2t^2)}{4} \end{aligned}$$

give the entries of $D_t = (d_{ij}(t))_{i,j=1}^2$:

$$\begin{aligned} d_{11}(t) &= 3(1 - e^{-2t}) + \beta^2(1 - e^{-2t}(1 + 2t + 3t^2)), \\ d_{12}(t) &= d_{21}(t) = \beta(1 - e^{-2t}(1 + 3t)), & d_{22}(t) &= 3(1 - e^{-2t}). \end{aligned}$$

In particular, in accordance with (4.2),

$$D_t = t \Lambda_C + O(t^2) \quad \text{as } t \downarrow 0, \quad \Lambda_C = 4I - (A_\beta + A_\beta^\top) = \begin{pmatrix} 6 & -\beta \\ -\beta & 6 \end{pmatrix}.$$

Furthermore,

$$\Sigma = \begin{pmatrix} \frac{1}{2} + \frac{\beta^2}{4} & \frac{\beta}{4} \\ \frac{\beta}{4} & \frac{1}{2} \end{pmatrix}, \quad P_C = I + 4\Sigma = \begin{pmatrix} 3 + \beta^2 & \beta \\ \beta & 3 \end{pmatrix},$$

and

$$A_\beta P_C + P_C A_\beta^\top = \begin{pmatrix} -6 & \beta \\ \beta & -6 \end{pmatrix}, \quad A_\beta(I + 2\Sigma) + (I + 2\Sigma)A_\beta^\top = \begin{pmatrix} -4 & \beta \\ \beta & -4 \end{pmatrix},$$

which recovers the thresholds 6 and 4 via Theorem 5.1(b).

Figure 1 shows the set of pairs (β, t) with $K^t H_I \subset H_I$ for $0 \leq \beta \leq 14$. For $6 < \beta \leq 14$ the boundary $t_1(\beta)$ was computed numerically: bisection on $\det D_t$ gives, for example, $t_1(7) \approx 0.3001$ and $t_1(8) \approx 0.3876$. For each of 320 values of β in $(6, 14]$, D_t is indefinite at the grid points $t < t_1(\beta)$ of a grid of $(0, 40]$ and positive semidefinite at the grid points $t \geq t_1(\beta)$. For $t \geq 40$, $D_t > 0$ holds rigorously, by the argument in the proof of Theorem 5.1(c). Indeed, $\|e^{A_\beta t}\| \leq (1 + \beta t)e^{-t}$, which is decreasing on $[1, \infty)$ and, for $\beta \leq 14$, below 10^{-14} at $t = 40$, while $\lambda_{\min}(P_C)/\lambda_{\max}(P_C) \geq 10^{-2}$ for $\beta \leq 14$. Hence, up to the grid resolution, $T_I = \{0\} \cup [t_1(\beta), \infty)$ for $6 < \beta \leq 14$.

Example 6.2 (Scalar noise). Let $d = 2$, $m = 1$, $C = I_2$, and

$$A = \begin{pmatrix} \frac{3}{2} & -2 \\ 2 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Then $\text{Tr } A = -\frac{1}{2}$ and $\det A = 1$, so the eigenvalues $-\frac{1}{4} \pm \frac{\sqrt{15}}{4}i$ have negative real parts and A is Hurwitz. Moreover $AB = (\frac{3}{2}, 2)^\top$, so $\det[B, AB] = 2 \neq 0$ and (A, B) is controllable. Since $\frac{1}{2}(A + A^\top) = \text{diag}(\frac{3}{2}, -2)$ and $BB^\top = \text{diag}(1, 0)$, we have

$$\begin{aligned} BB^\top - \frac{1}{2}(A + A^\top) &= \text{diag}(-\frac{1}{2}, 2) \not\geq 0, \\ 2BB^\top - \frac{1}{2}(A + A^\top) &= \text{diag}(\frac{1}{2}, 2) \geq 0. \end{aligned}$$

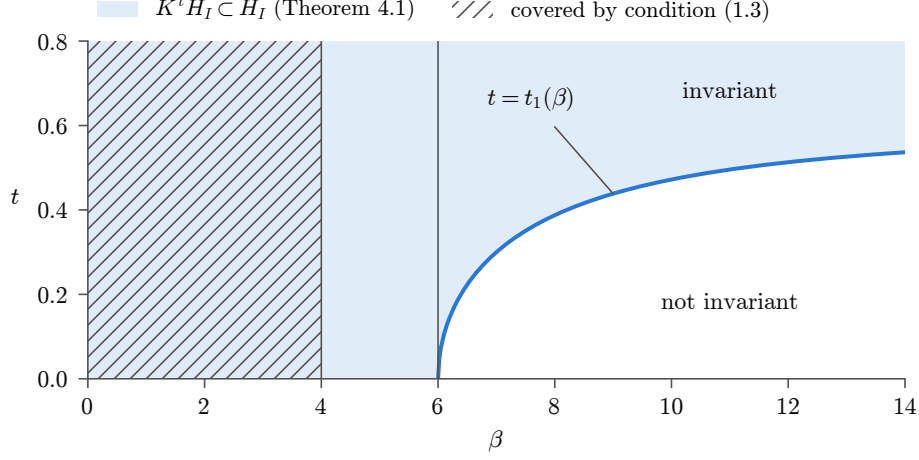


FIGURE 1. Admissible times for Example 6.1 ($A = A_\beta$, $B = C = I_2$). Shaded: pairs (β, t) with $K^t H_I \subset H_I$, determined by the criterion $D_t \geq 0$ of Theorem 4.1. Hatched: pairs covered by the sufficient condition (1.3) of [12], i.e. $\beta \leq 4$. For $\beta \leq 6$ invariance holds for all $t \geq 0$ (Theorem 4.4). For $6 < \beta \leq 14$ it fails for $0 < t < t_1(\beta)$ and holds for $t \geq t_1(\beta)$. The curve $t_1(\beta)$ was computed numerically (Example 6.1 and Appendix A).

So (1.3) fails and (1.5) holds, and H_I is invariant under every K^t although the noise is one-dimensional.

7. FURTHER REMARKS

Lemma 7.1 (Inner products of Gaussian bumps). *Let C', C'' be symmetric positive definite and $y, y' \in \mathbb{R}^d$. Then $k_y^{C'} \in H_{C''}$ if and only if $G := 2C'^2 - C''^2$ is positive definite. In that case*

$$\langle k_y^{C'}, k_{y'}^{C'} \rangle_{C''} = \frac{(\det C')^2}{\det C'' (\det G)^{1/2}} \exp(-(y - y')^\top G^{-1} (y - y')).$$

Proof. By (2.3), $|\widehat{k_y^{C'}}|^2 w_{C''} = \kappa_{C'}^2 \exp(-\frac{1}{4} \xi^\top G \xi)$, which is integrable if and only if $G > 0$. If $G > 0$, then (2.5) and (2.1) with $P = \frac{1}{4} G$ give

$$\begin{aligned} \langle k_y^{C'}, k_{y'}^{C'} \rangle_{C''} &= (2\pi)^{-d} \frac{\kappa_{C'}^2}{\kappa_{C''}} \int e^{-\frac{1}{4} \xi^\top G \xi - i\xi \cdot (y - y')} d\xi \\ &= (2\pi)^{-d} \frac{\pi^d (\det C')^2}{\pi^{d/2} \det C''} \pi^{d/2} 2^d (\det G)^{-1/2} e^{-(y - y')^\top G^{-1} (y - y')}, \end{aligned}$$

and $(2\pi)^{-d} \pi^d 2^d = 1$. $\square$

Remark 7.2 (Why kernel expansions cannot detect non-invariance). By (3.2) and (2.3),

$$\widehat{K^t k_z^C}(\xi) = e^{-t \operatorname{Tr} A} \kappa_C e^{-i\xi \cdot e^{-At} z} w_{C_t}(\xi)^{-1}, \quad \xi \in \mathbb{R}^d,$$

and since $\det C_t = e^{-t \operatorname{Tr} A} \det(C^2 + 2\Sigma(t))^{1/2}$, this means $K^t k_z^C = \tau_t k_{e^{-At} z}^{C_t}$, as in the proof of [12, Proposition 4.1]. By Lemma 7.1, $K^t k_z^C \in H_C$ if and only if $2C_t^2 - C^2 > 0$. Now $2C_t^2 - C^2 = e^{-At}(C^2 + D_t)e^{-A^\top t}$ and $D_t \rightarrow 0$ as $t \downarrow 0$. Hence, for all sufficiently small t , K^t maps $\operatorname{span}\{k_z^C : z \in \mathbb{R}^d\}$ into H_C regardless of the sign of D_t . Indeed, a finite kernel expansion f satisfies $|\widehat{f}|^2 w_C \leq c w_C^{-1}$ for some $c \geq 0$ by (2.3), so the right-hand side of (3.4) is finite whenever $C^2 + D_t > 0$. Witnesses of non-invariance must therefore have Fourier transforms that decay more slowly; those of Lemma 2.2 decay, within a cone of directions, as slowly as membership in H_C permits. This explains why the necessary condition $\operatorname{span}\{k_x^{C_t} : x \in \mathbb{R}^d\} \subset H_C$ noted in [12, Remark 4.2] is too weak to decide the question.

Remark 7.3 (Consistency with [12, Proposition 4.1]). By Corollaries 4.3 and 2.3, $H_{\tilde{C}_t} \subset H_{C_t}$ with $\|g\|_{\tilde{C}_t}^2 \leq (\det \tilde{C}_t / \det C_t) \|g\|_{C_t}^2$. Together with Theorem 4.2 this gives

$$\begin{aligned} \|K^t f\|_{C_t}^2 &\leq \frac{\det \tilde{C}_t}{\det C_t} \tilde{\tau}_t \|f\|_{\tilde{C}}^2 \\ &= \frac{\det(C^2 + 4\Sigma(t))^{1/2}}{\det(C^2 + 2\Sigma(t))^{1/2}} \cdot \frac{\det C}{\det(C^2 + 4\Sigma(t))^{1/2}} \|f\|_C^2 = \tau_t \|f\|_C^2, \end{aligned}$$

which is exactly the bound of [12, Proposition 4.1]. Thus Theorem 4.2 refines that proposition, and the only loss in the argument of [12] is the inclusion $H_{\tilde{C}_t} \subsetneq H_{C_t}$. On kernel sections, Theorem 4.2 states that

$$\langle K^t k_z^C, K^t k_{z'}^C \rangle_{\tilde{C}_t} = \tilde{\tau}_t k^C(z, z'), \quad z, z' \in \mathbb{R}^d.$$

This identity also follows directly from $K^t k_z^C = \tau_t k_{e^{-At} z}^{C_t}$ (Remark 7.2) and Lemma 7.1, because $2C_t^2 - \tilde{C}_t^2 = e^{-At} C^2 e^{-A^\top t}$.

Remark 7.4 (Deterministic linear flows). For $B = 0$ we have $\Sigma(t) = 0$ and $K^t f = f \circ e^{At}$. The proof of Theorem 4.1 works verbatim with e^{At} replaced by an arbitrary invertible matrix E . It shows that the composition operator $f \mapsto f \circ E$ maps H_C into itself if and only if $EC^2 E^\top \leq C^2$, i.e. $\|C^{-1} E C\| \leq 1$, and that its norm is then $|\det E|^{-1/2}$. Ikeda, Ishikawa and Sawano [4] proved that on a large class of RKHSs of analytic positive definite functions only affine maps induce bounded composition operators. For the Fock space of entire functions, bounded composition operators were characterized in [2]; there, too, the symbol must be an affine map whose linear part is a contraction.

We finally record what the method gives for other translation-invariant kernels. Let $\varphi: \mathbb{R}^d \rightarrow \mathbb{R}$ be continuous, integrable and positive definite,

with $\widehat{\varphi}$ continuous, integrable and strictly positive. Let H_φ denote the RKHS of $(x, y) \mapsto \varphi(x - y)$. By [16, Theorem 10.12], $H_\varphi = \{f \in C(\mathbb{R}^d) \cap L^2(\mathbb{R}^d) \text{ real-valued} : \int |\widehat{f}|^2 / \widehat{\varphi} < \infty\}$, with $\langle f, g \rangle_{H_\varphi} = (2\pi)^{-d} \int \widehat{f} \widehat{g} / \widehat{\varphi}$ in our normalization of the Fourier transform. For $\varphi(z) = \exp(-z^\top C^{-2} z)$ this is Lemma 2.1.

Proposition 7.5. *Let φ be as above and $t \geq 0$, and put*

$$m_t(\eta) := \frac{\widehat{\varphi}(\eta) \exp(-\eta^\top \Sigma(t) \eta)}{\widehat{\varphi}(e^{A^\top t} \eta)}, \quad \eta \in \mathbb{R}^d.$$

Then $K^t H_\varphi \subset H_\varphi$ if and only if $M_t := \sup_\eta m_t(\eta)$ is finite, and in this case

$$\|K^t\|_{H_\varphi \rightarrow H_\varphi}^2 = e^{-t \operatorname{Tr} A} M_t.$$

For the Gaussian kernel, i.e. for $\varphi(z) = e^{-z^\top C^{-2} z}$, we have $m_t(\eta) = e^{-\frac{1}{4} \eta^\top D t \eta}$; this recovers Theorem 4.1(a),(b).

Proof. Elements of H_φ are bounded, continuous and in L^2 , so Lemma 3.2 applies. The computation in the proof of Proposition 3.3 gives

$$\|K^t f\|_{H_\varphi}^2 = (2\pi)^{-d} e^{-t \operatorname{Tr} A} \int |\widehat{f}(\eta)|^2 \frac{m_t(\eta)}{\widehat{\varphi}(\eta)} d\eta$$

whenever the right-hand side is finite, and $K^t f \notin H_\varphi$ otherwise. If $M_t < \infty$, then $\|K^t f\|_{H_\varphi}^2 \leq e^{-t \operatorname{Tr} A} M_t \|f\|_{H_\varphi}^2$. For the reverse inequality, let $\delta > 0$ and choose η_0 with $m_t(\eta_0) > M_t - \delta$. The function m_t is continuous and even. Hence, for small $r > 0$, the function $f_r := (\mathbf{1}_{U_r})^\vee$ with $U_r := \{|\eta - \eta_0| < r\} \cup \{|\eta + \eta_0| < r\}$ belongs to $H_\varphi \setminus \{0\}$ and satisfies $\|K^t f_r\|^2 \geq e^{-t \operatorname{Tr} A} (M_t - \delta) \|f_r\|^2$.

Suppose now that $M_t = \infty$. For $k \in \mathbb{N}$ the set $\{m_t > 16^k\}$ is open, nonempty and symmetric. Choose a bounded symmetric set E_k of positive measure inside it, put $a_k := (\int_{E_k} \widehat{\varphi}^{-1})^{-1/2}$ and $F := \sum_k 2^{-k} a_k \mathbf{1}_{E_k}$. By Minkowski's inequality, $(\int F^2 / \widehat{\varphi})^{1/2} \leq \sum_k 2^{-k} = 1$. Since $\widehat{\varphi} \in L^1$, the Cauchy–Schwarz inequality gives $F \in L^1$. Also $F \in L^2$, because $\widehat{\varphi}$ is bounded by $\|\varphi\|_{L^1}$. As F is real and even, $f := F^\vee$ belongs to H_φ . Since $F \geq 2^{-k} a_k \mathbf{1}_{E_k}$ for every k , $\int F^2 m_t / \widehat{\varphi} \geq 4^{-k} a_k^2 16^k \int_{E_k} \widehat{\varphi}^{-1} = 4^k$. Hence $K^t f \notin H_\varphi$. The Gaussian case follows from (2.3) and a direct computation. $\square$

Example 7.6 (Sobolev spaces). For $s > d/2$ let $\widehat{\varphi}(\xi) = (1 + |\xi|^2)^{-s}$. This is the Matérn kernel, and H_φ is the space of real-valued functions in the Sobolev space $H^s(\mathbb{R}^d)$, with the norm $((2\pi)^{-d} \int |\widehat{f}|^2 (1 + |\xi|^2)^s)^{1/2}$. Since $(1 + |e^{A^\top t} \eta|^2) / (1 + |\eta|^2) \leq \max\{1, \|e^{At}\|^2\}$ and $\exp(-\eta^\top \Sigma(t) \eta) \leq 1$, we get $\sup m_t \leq \max\{1, \|e^{At}\|^2\}^s < \infty$. By Proposition 7.5, $H^s(\mathbb{R}^d)$ is invariant under every K^t , for all A and B , and $\|K^t\| \leq e^{-t \operatorname{Tr} A/2} \max\{1, \|e^{At}\|\}^s$. The restriction (1.5) therefore comes from the Gaussian decay of the spectral density: for the Gaussian kernel the ratio $\widehat{\varphi} / (\widehat{\varphi} \circ e^{A^\top t})$ in m_t is the exponential of

a quadratic form, which competes with the Gaussian factor $\exp(-\eta^\top \Sigma(t) \eta)$. Koopman invariance of the native spaces of Wendland kernels, which are norm-equivalent to Sobolev spaces, is proved in [6]; stochastic dynamics are treated in [3].

Remark 7.7 (Outlook). The proofs use that, for (1.1), K^t acts on Fourier transforms by composition with a linear map followed by multiplication with a Gaussian. For state-dependent diffusion coefficients or nonlinear drifts this structure is lost, and we do not know an analogous characterization of Koopman-invariant Gaussian RKHSs in that generality.

APPENDIX A. NUMERICAL CHECKS

The identities above were also checked with three Python scripts, which are provided as supplementary material: two in floating-point arithmetic (NumPy, SciPy and Matplotlib; random test data from a fixed seed) and one in exact arithmetic (SymPy). No proof in this paper depends on these checks. Matrix errors are measured in the entrywise norm $|M|_{\max} := \max_{i,j} |M_{ij}|$. The covariance $\Sigma(t)$ was computed with Van Loan's block-exponential formula [14] on steps of length at most 0.25, combined with the recursion $\Sigma(s+h) = \Sigma(h) + e^{Ah}\Sigma(s)e^{A^\top h}$. The one-step formula cancels catastrophically for large t .

- (1) $\Sigma(t)$ agrees with its value $\Sigma_q(t)$ from adaptive quadrature: $|\Sigma(t) - \Sigma_q(t)|_{\max} \leq 2.7 \times 10^{-13} |\Sigma_q(t)|_{\max}$ (random $A \in \mathbb{R}^{3 \times 3}$, $B \in \mathbb{R}^{3 \times 2}$, $t \in \{0.3, 1.7, 4\}$).
- (2) Let R_t be the right-hand side of (4.2), computed by adaptive quadrature. Then $|D_t - R_t|_{\max} \leq 5.2 \times 10^{-13} \max\{1, |R_t|_{\max}\}$ for three random triples (A, B, C) with A not Hurwitz (spectral abscissae 1.645, 1.556, 2.043) and $t \in \{0.05, 0.8, 2.5\}$. Moreover $|t^{-1}D_t - \Lambda_C|_{\max} \leq 3.7 \times 10^{-6} \max\{1, |\Lambda_C|_{\max}\}$ at $t = 10^{-6}$, consistent with the $O(t)$ remainder.
- (3) Theorem 5.1(a),(b) for a random Hurwitz $A \in \mathbb{R}^{3 \times 3}$ and $B \in \mathbb{R}^{3 \times 1}$: the entries of $D_t - (P_C - e^{At}P_Ce^{A^\top t})$ are at most 2.4×10^{-14} in modulus for $t \in \{0.1, 1, 6\}$, and those of $AP_C + P_CA^\top + \Lambda_C$ at most 1.8×10^{-14} .
- (4) Theorem 4.2 on kernel sections, for random 2×2 data and $t \in \{0.2, 0.9\}$. We compute the Gram matrix $(\langle K^t k_{z_i}^C, K^t k_{z_j}^C \rangle_{\tilde{C}_t})_{i,j \leq 4}$ from Lemma 7.1 and the formula $K^t k_z^C = \tau_t k_{e^{-At}z}^{C_t}$ of [12, proof of Proposition 4.1]. Its entries differ from those of $\tilde{\tau}_t(k^C(z_i, z_j))_{i,j}$ by at most 3.8×10^{-14} . The value $\|K^t k_z^C\|_C^2 = e^{-t \operatorname{Tr} A} \det C \det(2C^2 + 4\Sigma(t) - e^{At}C^2e^{A^\top t})^{-1/2}$, obtained from (3.4), agrees with the value given by that formula and Lemma 7.1 to a relative error of 1.1×10^{-13} .
- (5) Monte Carlo check of the formula $(K^t k_z^C)(x) = \tau_t k_{e^{-At}z}^{C_t}(x)$ for Example 6.1 with $\beta = 5$, $t = 0.4$, $x = (0.3, -0.2)^\top$ and $z = (0.5, 0.4)^\top$,

using 2×10^6 exact Gaussian samples of Z_t : $0.436373 \pm 2.2 \times 10^{-4}$ (one standard error) against the closed form 0.436051, a deviation of 1.45 standard errors.

- (6) Example 6.1 for $\beta \in \{4, 5, 6, 7, 8\}$: (1.3) holds only for $\beta = 4$, and (1.5) exactly for $\beta \leq 6$. On 4200 grid points in $(0, 12]$, D_t has no negative eigenvalue for $\beta \leq 6$. For $\beta = 7$ and $\beta = 8$ the grid points with indefinite D_t form an initial segment ending at 0.298 and 0.385, respectively. For $\beta \in \{5, 6\}$ the matrix $C_t^2 - C^2$ of [12] has a negative eigenvalue (-1.2×10^{-2} , resp. -2.4×10^{-2}) at some $t \leq 0.0124$, although $D_t \geq 0$ there. The closed form of D_t in Example 6.1 matches the numerically computed D_t to within 2.5×10^{-14} .
- (7) Figure 1: for 320 values of $\beta \in (6, 14]$ the set $\{t \in (0, 40] : D_t \geq 0\}$, sampled on 2×10^5 grid points, is an interval $[t_1(\beta), 40]$; t_1 was located by bisection on $\det D_t$ and increases from 0.053 at $\beta = 6.025$ to 0.537 at $\beta = 14$. For $t \geq 40$ the condition of Theorem 5.1(c) holds.
- (8) Example 6.2: the matrix A has the eigenvalues $-0.25 \pm 0.968i$, and the controllability matrix $[B, AB]$ has rank 2. Moreover $\lambda_{\max}(\frac{1}{2}(A + A^\top) - BB^\top) = 0.5$ and $\lambda_{\max}(\frac{1}{2}(A + A^\top) - 2BB^\top) = -0.5$, and $\lambda_{\min}(D_t) \geq 0$ on 3000 grid points in $(0, 10]$.
- (9) Proposition 7.5, Gaussian case: for random $A \in \mathbb{R}^{3 \times 3}$, $B \in \mathbb{R}^{3 \times 2}$ and C , $t \in \{0.3, 1.2\}$, and 1000 random η , the quantities $\log m_t(\eta)$ and $-\frac{1}{4}\eta^\top D_t \eta$ agree to a relative error of 3.2×10^{-14} .
- (10) Example 7.6: for a random $A \in \mathbb{R}^{2 \times 2}$ that is not Hurwitz (spectral abscissa 0.857), $B \in \mathbb{R}^{2 \times 1}$, $s = 1.3$ and $t \in \{0.5, 2\}$, the bound $m_t(\eta) \leq \max\{1, \|e^{At}\|^2\}^s$ holds at 8.04×10^4 points η with $10^{-3} \leq |\eta| \leq 10^6$.

In total the script for items 1–6 and 8–10 performs 38 automated checks, all of which pass. The exact-arithmetic script confirms the closed forms in Examples 6.1 and 6.2 (the stationary covariance in Example 6.2 is $\Sigma = \begin{pmatrix} 5 & 4 \\ 4 & 4 \end{pmatrix}$), the constants in (2.1)–(2.5) and Lemma 7.1, the bound $\lambda_{\min}(P_C)/\lambda_{\max}(P_C) \geq 10^{-2}$ for $\beta \leq 14$ used in Example 6.1, and Theorem 4.2 for $d = 1$ and $f = k_z^C$. It performs 52 checks, all of which pass.

Data availability. No external data were used. The scripts that produce the numbers in Appendix A and Figure 1 are provided as supplementary material.

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